DEEP LEARNING

Project Presentation

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PART ONE

Transitioning from Neural Architectures to Advanced Transfer Learning

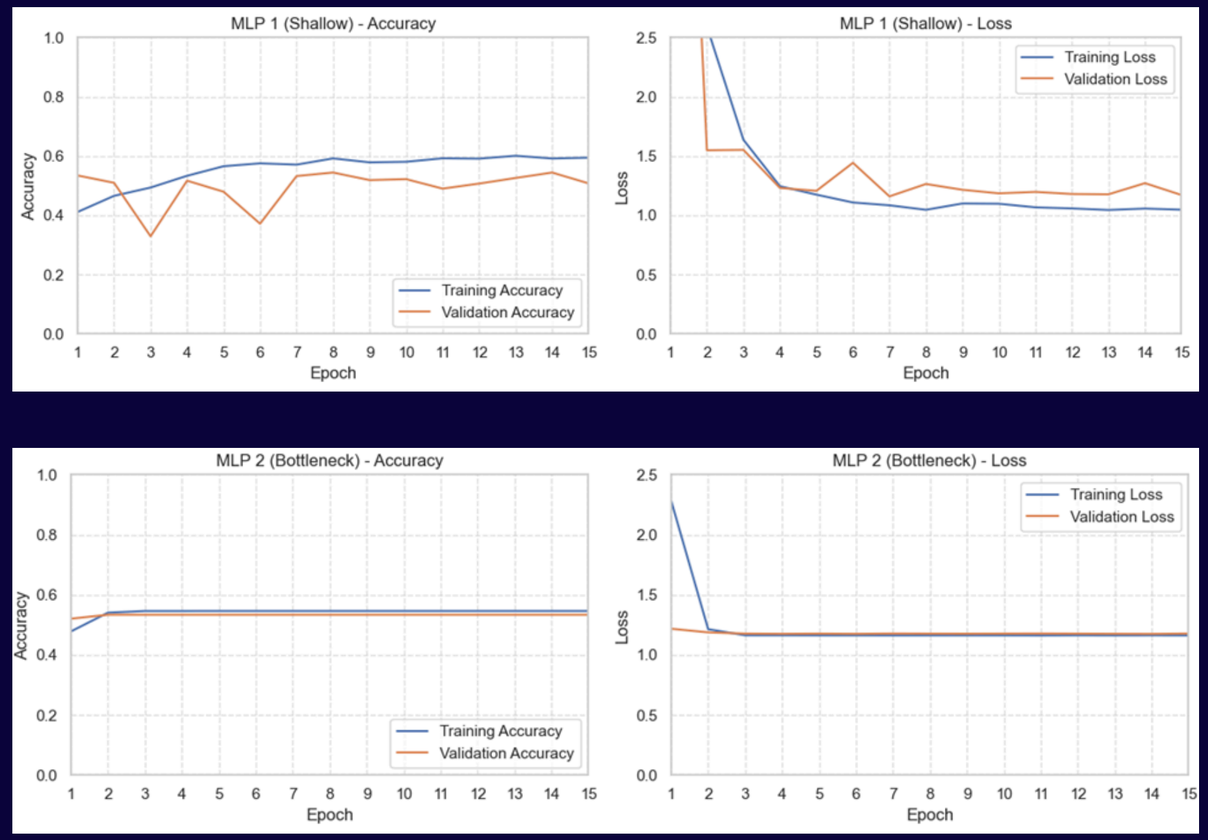
Spatial Awareness vs. Model Complexity

Dataset Overview
7,643 Flickr images across 5 distinct classes

The Verdict
MLP 2 generalized better due to narrower layers and Dropout regularization

The Problem
Flattening 2D images into 1D arrays destroys essential spatial hierarchies

Performance Cap
Validation accuracy was restricted to approximately 54%



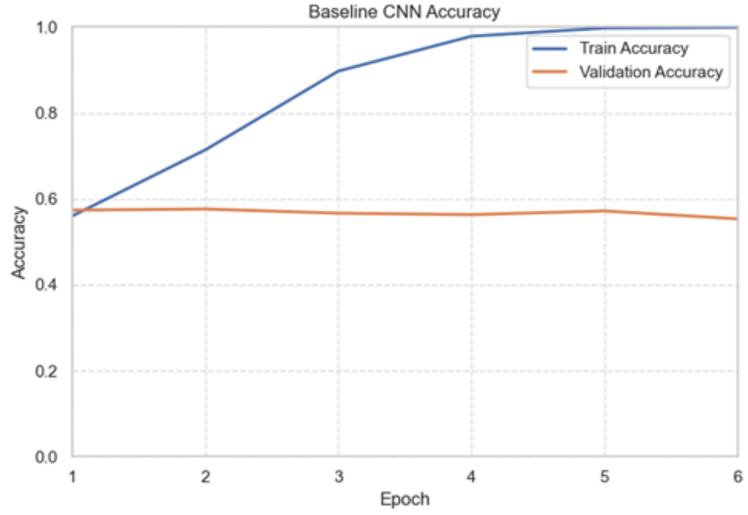
The CNN Bottleneck: Parameters & Overfitting

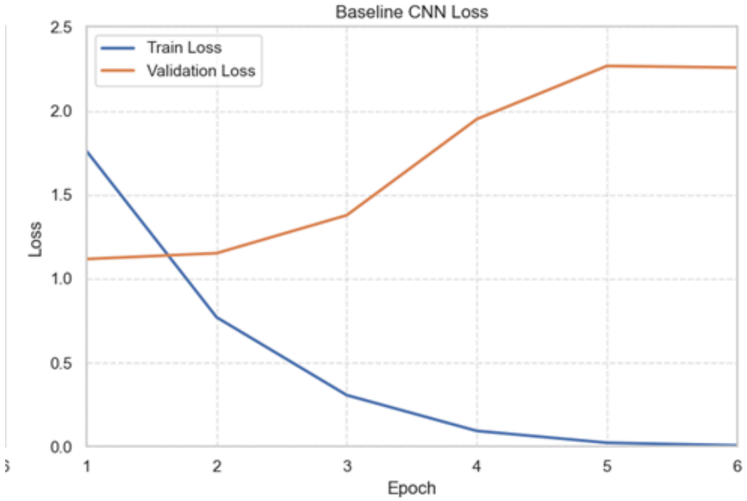
Baseline CNN
Introduced convolutional layers to extract spatial features

The Bottleneck
Using a Flatten layer after convolution created a massive 125 million parameter count

Observation
Training accuracy reached nearly 100%, but validation accuracy declined steadily

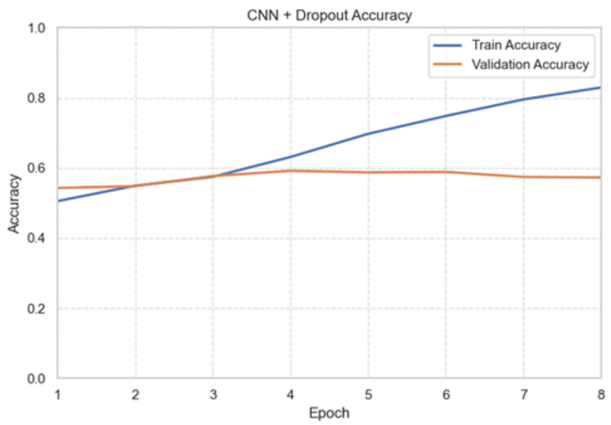
Conclusion
Massive parameter counts lead to memorization (overfitting) rather than true generalization

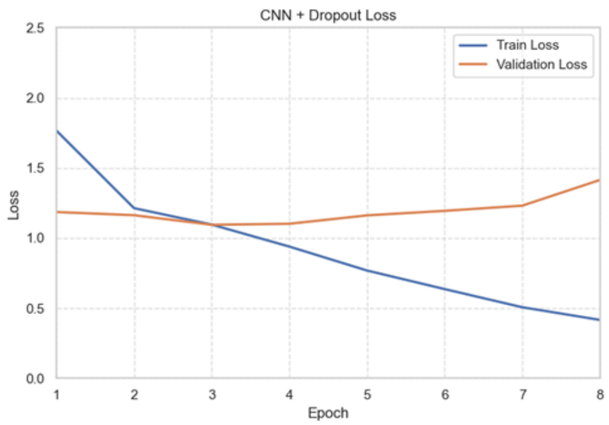




Solving the Bottleneck: Batch Norm & Dropout

Experiment 1– CNN + DROUPOUT

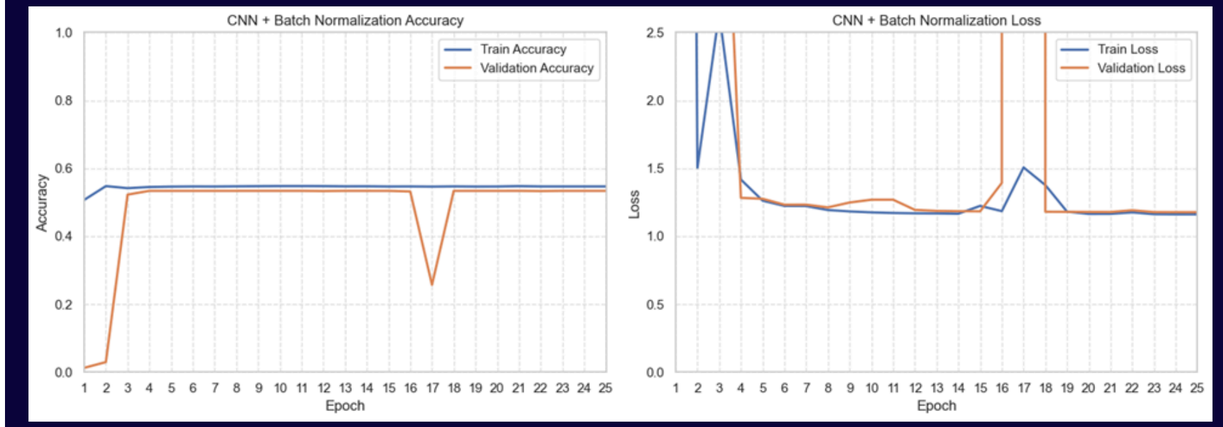




- **Optimization** Added a 50% Dropout layer to curb the baseline's severe overfitting.
- **Impact:** Successfully narrowed the gap between training and validation metrics.
- **Early Stopping:** Halted at epoch 8; peak performance occurred at epoch 3.
- **Results:** Validation accuracy reached 57.68% (loss: 1.09).
- **Core Flaw:** Dropout alone could not resolve the 125M+ parameter bottleneck caused by the Flatten layer.

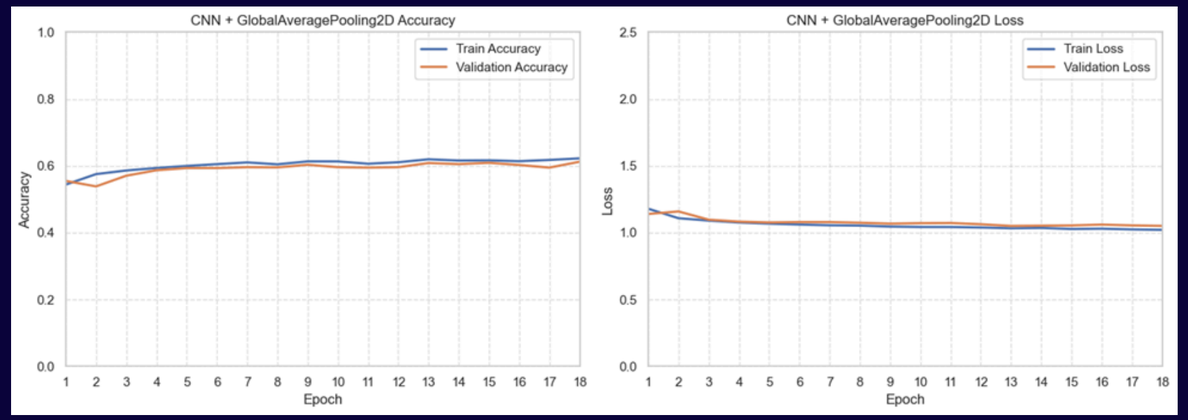
Experiment 2 – Batch Normalization

- **Goal:** Stabilize and accelerate learning by normalizing activations within the convolutional layers.
- **Architecture:** Inserted two Batch Normalization layers post-activation.
- **Impact:** Successfully neutralized extreme overfitting, keeping training and validation metrics closely aligned.
- **Trade-off:** While stabilized, it struggled to learn complex patterns, with accuracy dropping to 53.37%.
- **Persistent Flaw:** The model still contained over 125 million parameters due to the Flatten layer.



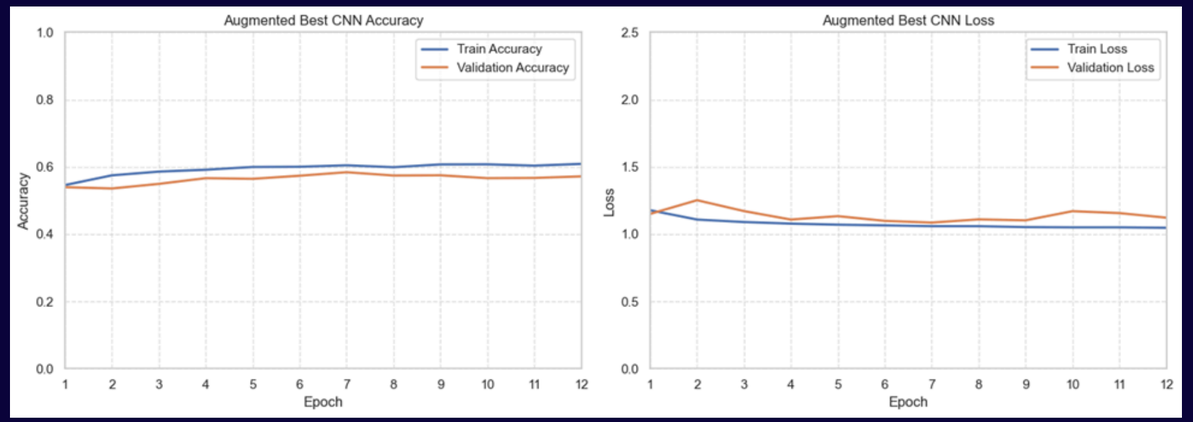
Solving the Bottleneck: Gobal avg pooling 2D

Experiment 3 – Global Average Pooling



- **Architectural Fix** Replaced the Flatten layer with GlobalAveragePooling2D (GAP2D) to average feature maps into single values.
- **99.9% Reduction** in trainable parameters from **125 Million to just 28,357.**
- **Generalization -** Drastically decreased overfitting compared to the baseline, forcing the model to learn more robust features.
- **Result-** Achieved a final validation accuracy of 61.00%.

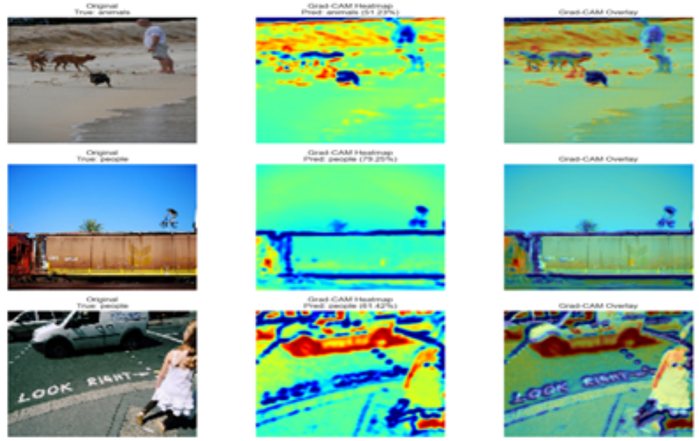
Data Augmentation & Robustness



- **Processing-** Integrated RandomFlip, RandomRotation, and RandomZoom layers directly into the model.
- **Goal** is to Increase the variety of training data without manually augmenting the dataset.
- **Augmented Performance:** Maintained high efficiency with a final validation accuracy of 58.40%.
- **Interpretability Insight:** Grad-CAM and Occlusion tests showed that while accurate, the model still relies on "environmental shortcuts".

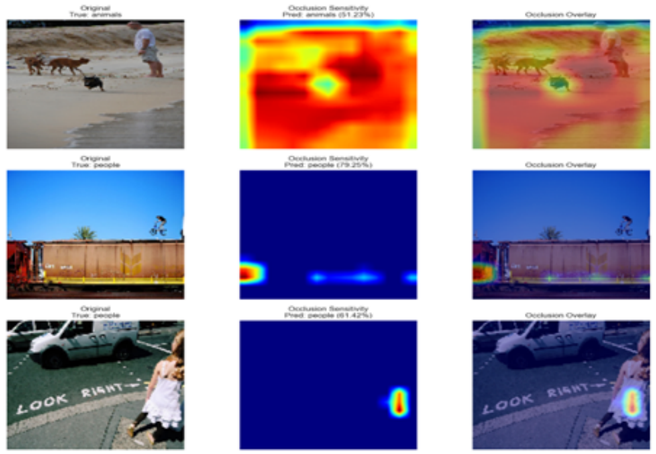
Solving the Bottleneck: Gobal avg pooling 2D

Grad-CAM Analysis



- **Heatmap Generation:** Highlights specific image regions that most strongly influence the model's classification.
- **Dispersed Attention:** The network's focus rarely isolates the true target subject, often bleeding into surroundings.
- **Shortcut Learning:** The model likely relies on background environmental cues rather than definitive visual traits.
- **Resolution Limits:** Computed from the final convolutional layer, providing a rough area of importance rather than pixel-perfect detail.

Occlusion Sensitivity



- **Perturbation Method:** Systematically masks image areas to measure changes in prediction confidence.
- **Heatmap Logic:** "Hot" colors indicate critical regions; "Cool" colors show regions with little effect on the prediction.
- **Irrelevant Highlighting:** In some samples, the model placed intense importance on small, irrelevant areas, completely missing the actual subject.
- **Contextual Reliance:** Confirms heavy dependence on environmental shortcuts to infer the class.

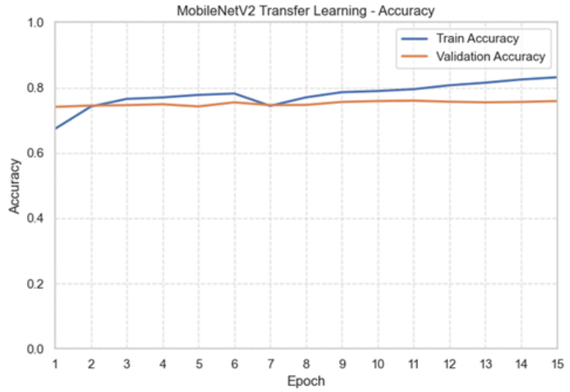
The Champion – MobileNetV2 & Final Comparison

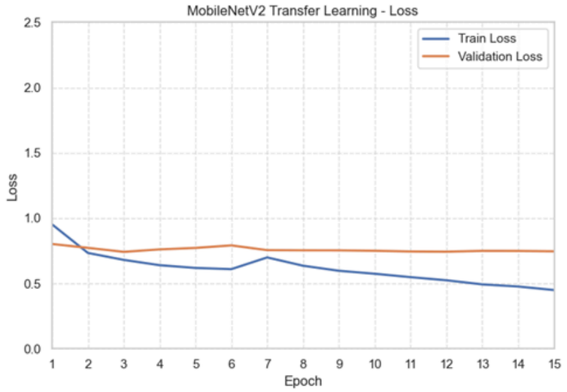
Strategic Advantage:
Leveraged a pre-trained backbone to extract robust visual features like edges, textures, and shapes.

Architecture:
Built a custom head with GlobalAveragePooling2D and Dropout (0.2) on top of the MobileNetV2 base.

Initial Training:
Achieved 74.62% validation accuracy with base weights frozen.

Fine-Tuning:
Unfreezing the base model yielded the project's peak performance of 75.67% validation accuracy.





Spatial Awareness is Critical: Moving from MLPs to CNNs proved that preserving image hierarchy is essential for classification.

Leveraging MobileNetV2 (ImageNet features) provided a highly optimized starting point that custom architectures could not match.

PART TWO

Transitioning from Neural Architectures to Advanced Transfer Learning

Project Overview & Objective

OBJECTIVE

To determine if qualitative "Risk Factors" (Item 1A) in SEC 10-K filings can predict a firm's stock performance over the subsequent 12 months

DATASET

Scraped from the web 20 years of data (2005–2025) covering over 9,000 filings from S&P 500 constituents using sec_edgar_downloader

KEY CHALLENGES

Bridging the gap between dense, unstructured narrative data and binary market outcomes (Stock Up vs. Stock Down)

TARGET DEFINITION

$$\text{Return}_{1Y} = \frac{\text{Price}_{END} - \text{Price}_{START}}{\text{Price}_{START}}$$

Y=1(Positive Performance): Assigned if the 12-month return was greater than zero.
Y=0 (Negative/Flat Performance): Assigned if the 12-month return was less than or equal to zero.

Data Acquisition & Engineering Pipeline

Scaling Qualitative Extraction Across 20 Years of Market History

Automated Web Scraping: Developed a high-velocity engine to dynamically retrieve S&P 500 constituents and fetch longitudinal Form 10-K filings.

Massive Corpus Scale: Successfully processed over 9,165 primary documents covering a two-decade horizon from 2005 to 2025.

Targeted "Item 1A" Extraction: Utilized BeautifulSoup and Regular Expressions (Regex) to isolate the "Risk Factors" section, which is uniquely forward-looking compared to retrospective financial audits.

Decomposition: Programmatically deleted HTML tables, images, and SVG graphics to ensure models focused on narrative sequences rather than visual artifacts.

Quality Control: Implemented a 1,500-character minimum threshold to filter out "Table of Contents" fragments and unsuccessful extractions.

Temporal Synchronization: Paired each text sequence with historical price data via the Yahoo Finance API, calculating a standardized one-year forward return window starting April 1st to allow for "filing season" digestion.

Recurrent Neural Network Experiments

The Struggle for Sequential Memory in 3,000-Token Financial Prose

Experimental Framework

Benchmarked 13 distinct neural architectures to identify the optimal balance between feature extraction and sequential memory.

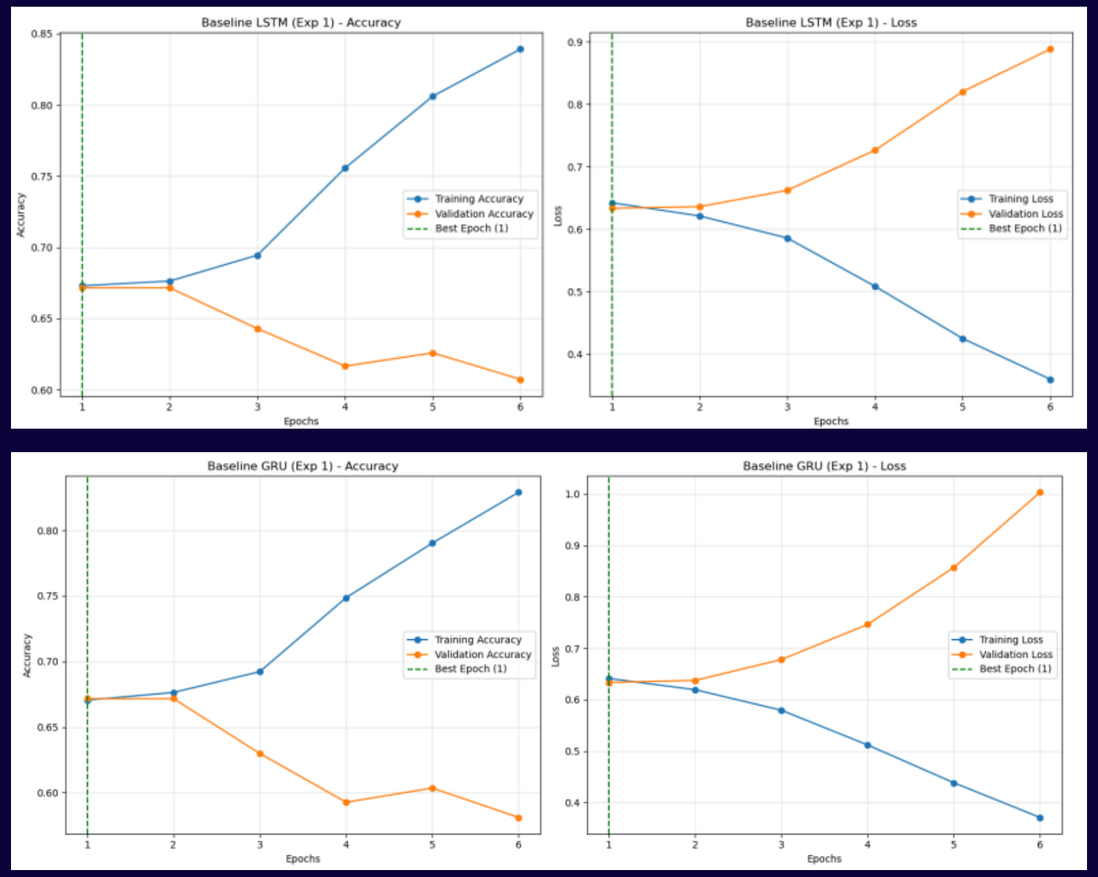
Architectural Logic

LSTM Gating Selectively filters 3,000-token sequences to "remember" structural risks while "forgetting" repetitive legal boilerplate.

GRU Efficiency Features 33% fewer parameters for faster convergence and higher sensitivity to concentrated "market downturn" language.

The Structural Variants

Stacked Layered units to create hierarchical representations
Bidirectional Processed text from both beginning-to-end and end-to-beginning.



Convolutional Neural Network Experiments

CNNs as "Keyword Spotters"

The Philosophy

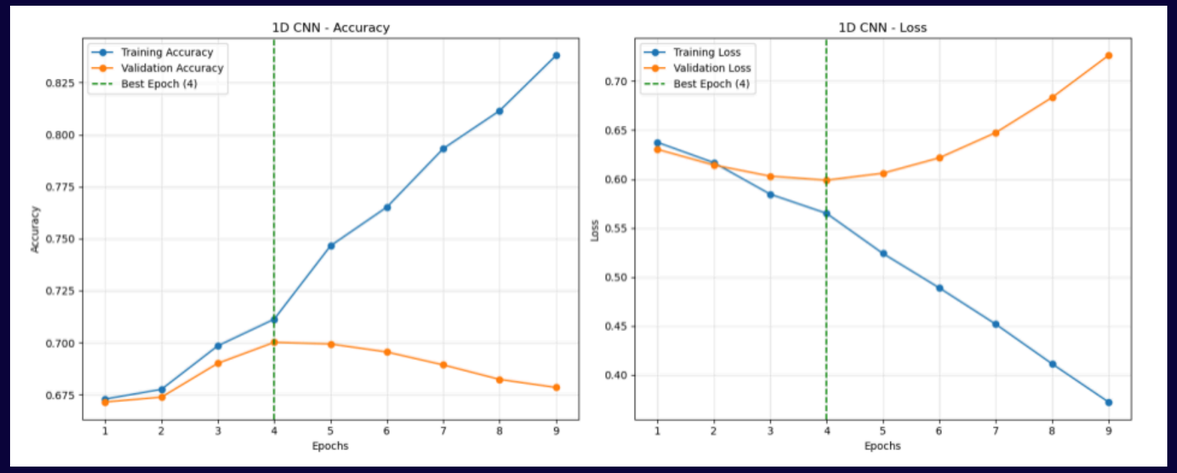
Instead of reading the whole sequence, CNNs use sliding windows (kernels) to identify high-impact "n-grams" or phrases.

Superior Performance

The Baseline CNN achieved **70.02%** validation accuracy, successfully breaking the **67%** baseline.

Architectural Advantage

Kernel Scanning: Employs 128 filters with a kernel size of 5 to scan for predictive 5-word phrases across the filing.
Global Max Pooling: Specifically designed to extract the single "most alarming" or "most positive" signal from the entire text.

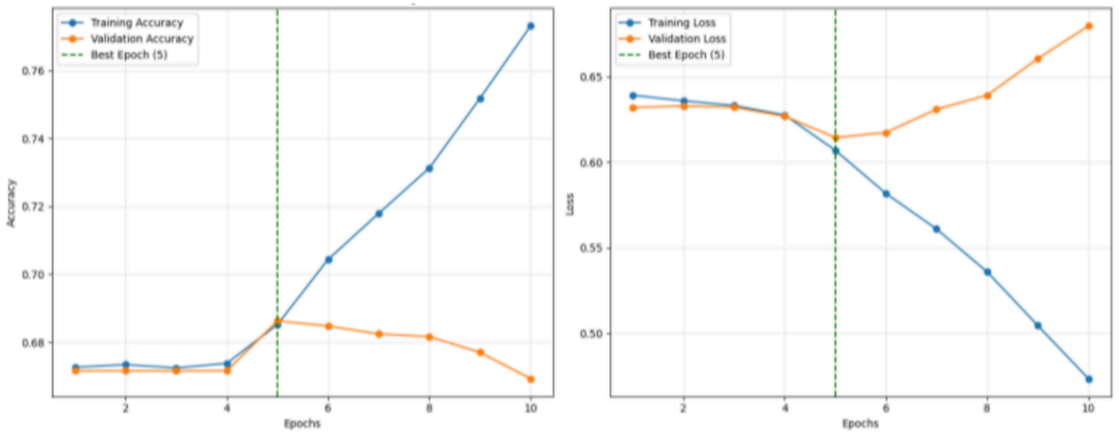


Key Insight: Signal vs. Noise

Keyword Spotting: CNNs act as "red flag" detectors by ignoring legal boilerplate to focus on high-activation phrases.
Feature Extraction: By identifying clusters like "supply chain disruption" or "substantial doubt," the model creates a robust link between textual sentiment and market volatility.

Attention Mechanisms and Transfer Learning

Attention-Enhanced Bi-GRU



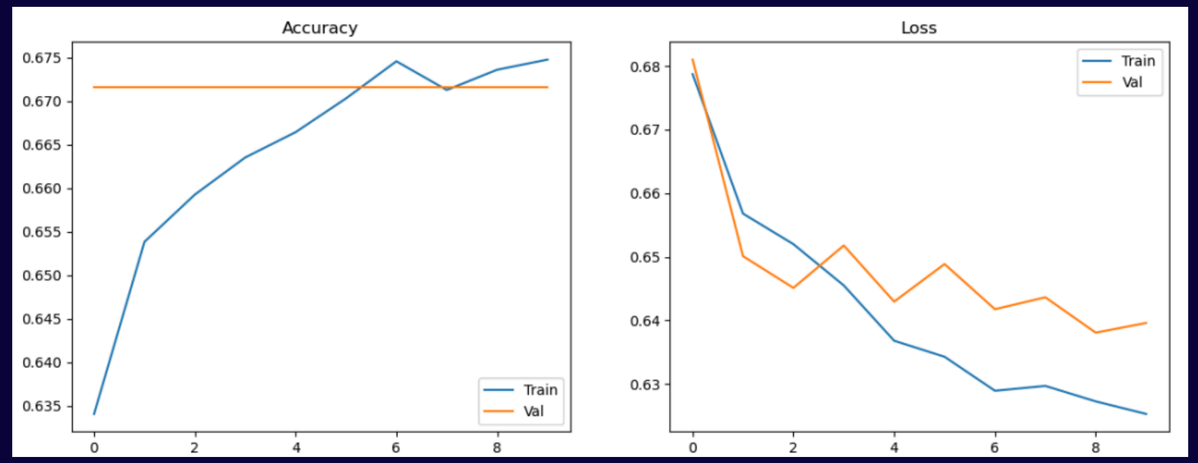
Mechanism: Integrates an Attention Layer to assign unique weights to every token, allowing the model to "highlight" high-impact phrases like "litigation" or "liquidity.

Result: Successfully broke the baseline trap to achieve 68.63% validation accuracy.

Hybrid FinBERT-Bi-GRU

Domain Intelligence: Leveraged FinBERT, pre-trained on financial corporate text to capture nuanced sentiment (e.g., the financial gravity of "liquid" vs. its general English meaning).

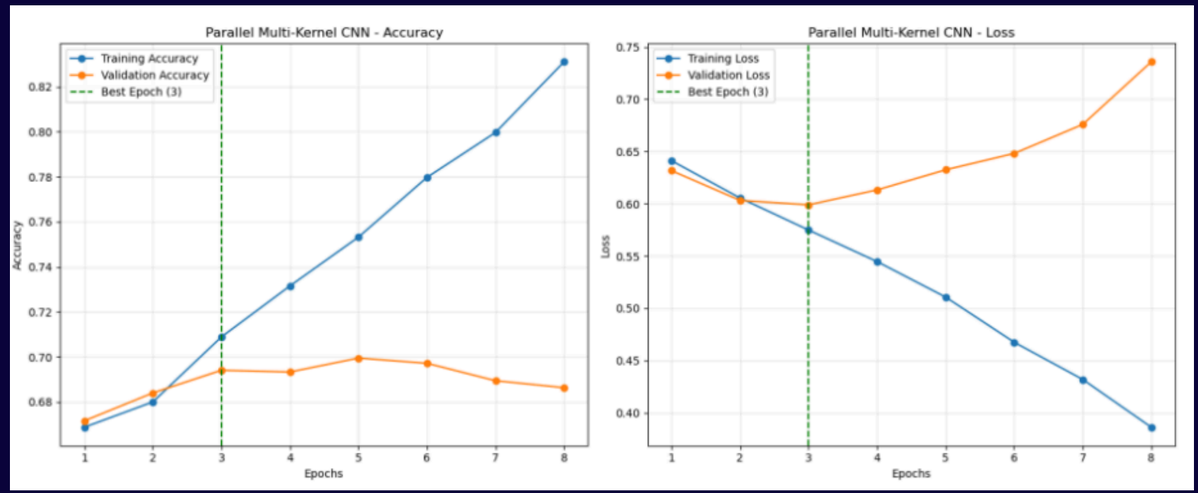
Stability: Demonstrated high resistance to the extreme overfitting seen in deep CNNs, providing the most stable loss reduction over 10 epochs.



The Architectural Winner

Parallel Multi-Kernel CNN (Robust Feature Synthesis)

While the **Baseline CNN hit a slightly higher accuracy** peak, the Parallel Multi-Kernel model is the **superior choice for real-world deployment due to** its "Multi-Scale" approach.



Uses three parallel branches (Kernel sizes 3, 4, and 5) to process the same embedding input simultaneously.

This recognizes that financial risks are not uniform; some are short "trigram" phrases, while others are complex 5-word semantic clusters.

Validation Accuracy: 69.40%.
Strategic Value: It provides the most robust "scanning" method for identifying critical risks within massive corporate datasets without sacrificing efficiency.

Conclusions & Future Outlook

THE TECHNICAL CEILING

persistent accuracy ceiling of ~70% was observed across 11 experiments.

CETERIS PARIBUS

External Shocks: "Black Swan" events (geopolitical conflicts, Fed policy shifts) can override textual sentiment.

TIMING

Efficient Markets: Per EMH, by the time a 10-K is filed, professional analysts have often already priced in the disclosed risks.

FUTURE IMPROVEMENTS

ransition to a Multimodal framework by merging qualitative text signals with "Hard Data" like Debt-to-Equity ratios, GDP growth, and the VIX "Fear Gauge.

Thank you,
QUESTIONS ?